

Medora: Consent-Governed Clinical AI Infrastructure for Bangladesh

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<https://medorahealth.vercel.app>

<https://github.com/CSE-3200-System-Project/Medora>

Abstract

A fluent language model can word clinical advice well, but fluency does not decide what patient data it should see or how much its output is allowed to do. Medora pairs a deployed clinical platform with a clear plan to test and prove that control layer in Bangladesh. Five named parts keep the questions apart: how far the AI may act (**Arohon**), whether the safety line held (**Lokkhon**), what consent costs in usefulness (**Shimana**), whether a fine-tuned model stays safe enough to deploy (**Maya**), and a national drug-identity list (**Akkhor**). Learned models handle speech, vision, and drafting; plain deterministic code decides what is disclosed and what takes effect. Today's evidence is 134 privacy cases (94.7% precision, 75.5% recall), 30 clinician-reviewed navigation fixtures (all seven emergencies kept on both scored paths), 12 source-grounded summaries, and an index of 7,389 drugs and 67,001 brands. Arohon's tiers are specified; Shimana, Maya, and a learned de-identifier are registered plans. Medora navigates and drafts for a clinician to approve. It does not diagnose, prescribe, counsel, or claim to be a medical device.

1. Problem and vision

Bangladesh needs useful digital health assistance without transferring clinical authority to a general-purpose model. Household out-of-pocket payments represented about 72.5% of current health expenditure in 2022. WHO's Bangladesh page reports that medicines and medical products accounted for 69.4% of out-of-pocket payments in the 2015 national accounts [1, 2]. WHO identifies health-model risks including inaccurate statements, automation bias, privacy loss, unequal access, and the delegation of difficult choices to a model [3]. These constraints require a fluent Bengali response to be evaluated alongside enforceable controls.

Bangladesh's new data regime turns that safety problem into an infrastructure problem. The Personal Data Protection Ordinance 2025 treats health information as sensitive personal data. It defines valid consent as informed, specific, voluntary, and affirmative. And it requires safeguards such as pseudonymisation, periodic risk assessment, testing, and audit [4]. Most provisions took immediate effect; selected institutional and enforcement provisions commence on a later gazetted date after eighteen months. Medora is an engineering tool for conformance, not a legal certificate; formal certification stays with accredited bodies. What it adds is to make consent status and data-sharing limits visible enough to test. The need is already official: in 2026 the Cabinet Division warned every ministry, and Bangladesh Bank warned every bank, against sending confidential data to third-party AI tools [20, 21]. Medora's answer is a reusable privacy layer, not a single application. Policy-before-inference, deny-by-default consent, redaction, pseudonymisation, local-first routing, and audit are domain-agnostic components that keep sensitive data inside the boundary and minimise what any external model can see. We build and prove them in health, the hardest setting, as a reference implementation for a privacy-aware, secure, AI-native Bangladesh whose same layer extends to any sector that holds sensitive records.

The gap is authority you can enforce in code. A safe system needs the power to take back a data-sharing grant, to hold the assistant to a fixed set of actions, to tie every answer to a supplied record, and to save a booking with its audit trail together or not at all. Medora therefore puts the policy check before the model and a human check after it. The thesis is short: *clinical AI should earn authority by risk class, consent, and evidence, not inherit it from model confidence.*

Objectives. Medora will (i) formalise tiered au-

thority, (ii) extend bilingual safety and privacy evaluation, (iii) quantify the utility cost of narrower consent, (iv) admit a locally deployable model only after red-flag and licence gates, and (v) keep drug identity openly reusable. Deployed, measured, specified, and planned work remain visually and verbally distinct.

2. Use cases and existing solutions

Medora serves patients, doctors, and infrastructure adopters. Patients use an installable bilingual client to hold records, manage sharing, inspect access history, find doctors, and book appointments. Doctors receive source-linked summaries and structured drafts that remain editable before effect. A hospital, research group, or health-software vendor can reuse the consent guard, benchmark runner, route registry, or medicine identity layer independently of the Medora interface.

These systems are complementary. OpenMRS, Bahmni, GNU Health, and the national SHR cover broader hospital workflows and standards-based interoperability. Medora contributes an inspectable boundary around assistive AI, coupled to Bangladesh medicine identity and reproducible containment tests.

3. The five-artifact framework

Each name maps to a technical function. Arohon (আরোহণ, “ascent”) specifies increasing authority; Lokkhon (লক্ষণ, “clinical sign”; a homophone evoking লক্ষণরেখা, the line not to cross) tests what the system must catch; Shimana (সীমানা, “frontier”) will measure utility against disclosure; Maya (ময়া, “illusion”) will test whether domain fluency preserves escalation sensitivity; and Akkhor (অক্ষর, “character”) supplies canonical drug identity. Akkhor canonicalises, Arohon governs, Lokkhon tests, Shimana prices, and Maya challenges the alternative.

3.1 Arohon: graded autonomy

Arohon sets how far the AI may go as a ladder of tiers, capped by how dangerous the case is. In the specified execution order, an endpoint declares an intended tier; the policy engine resolves actor, purpose, recipient, scopes, and risk class; the ceiling lowers the tier when needed; and the selected tier is logged with the correlation identifier. Current routes already provide deny-by-

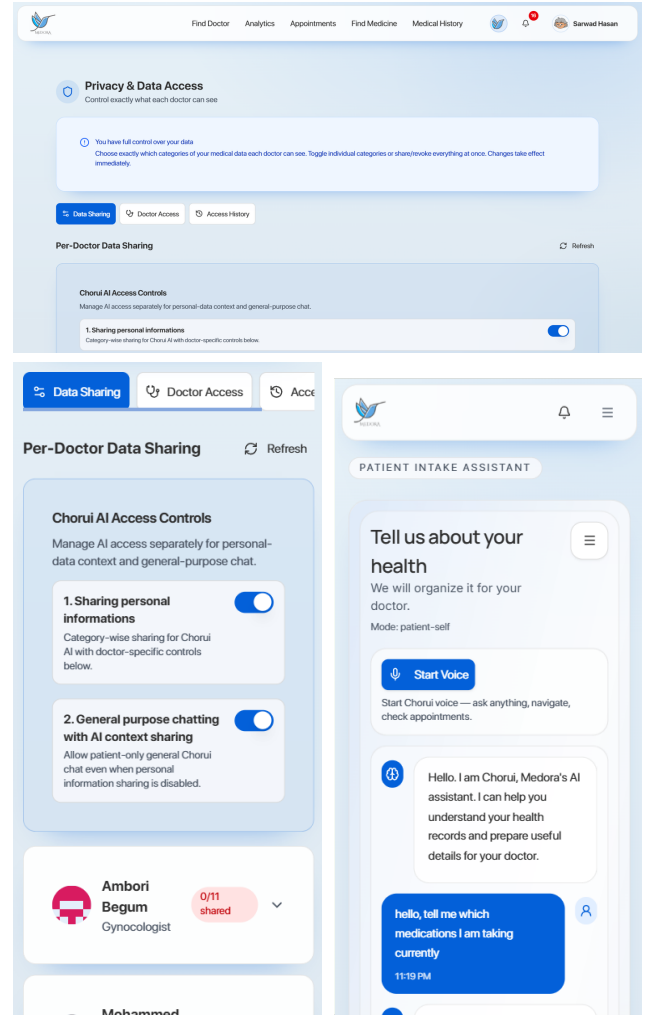


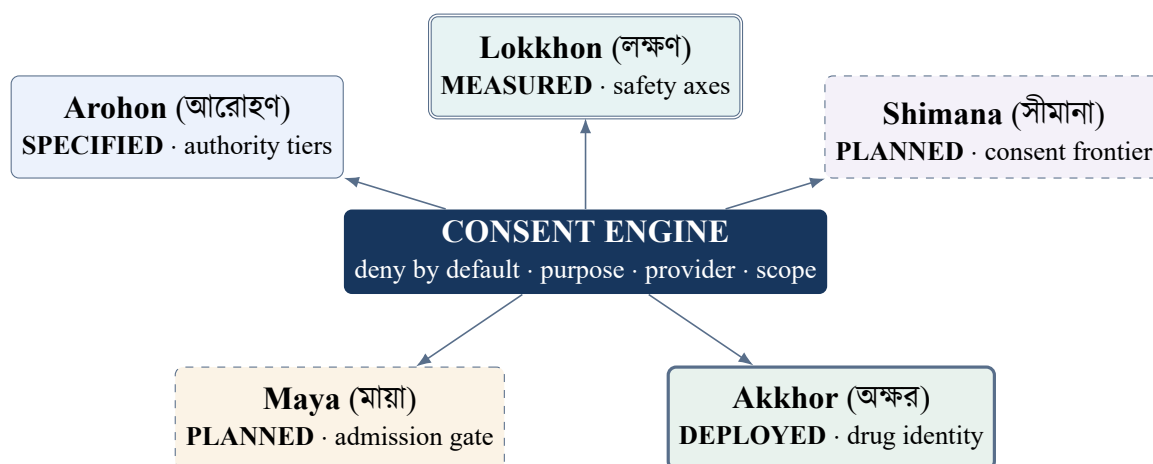
Figure 1. The deployed evidence surface: recipient/category sharing on desktop, the same controls at a phone viewport, and the mobile patient-intake assistant. These interfaces show that consent boundaries and assistive status are user-visible rather than policy prose alone.

default consent, deterministic red flags, source-linked drafts, and human confirmation; formal tier enums, ceilings, and tier logs are the specified next increment.

Ceilings encode risk-specific authority. Under the specification, routine queries remain at L1–L2 and out-of-scope inputs fall to L0. Cardiac, stroke, anaphylaxis, and obstetric red flags may surface an L3 emergency takeover, while L4 activates only with a previously granted and still-valid notification recipient. The self-harm path is capped at L3 in this proposal. It presents clinician-authored supportive text and time-aware human-help options, excludes method content, and keeps dispatch under human control. This policy is a deliberate design boundary for independent clinical and ethics review; it is an engineering specifica-

Table 1. Publicly documented scope of representative systems. “Outside reviewed scope” limits the comparison to properties established in cited public material.

System	Primary scope	Patient consent surface	Assistive-AI containment	Bangladesh drug identity	Public status
OpenMRS	Modular EMR platform	Deployment-specific	Outside reviewed core scope	Deployment-supplied dictionaries	Mature open source [5]
Bahmni	Hospital, laboratory, stock, billing	OpenMRS-based roles and workflow	Outside reviewed core scope	Deployment-supplied dictionaries	Open-source hospital suite [6]
GNU Health	EMR, HMIS, public health, FHIR	Role/access facilities	Outside reviewed core scope	WHO essential medicines	Mature open source [7]
Bangladesh SHR	National HIE, Health ID, longitudinal record	Citizen-controlled record sharing	Assistive-AI layer outside reviewed scope	National terminology registries	Government infrastructure [8]
DocTime Praava	Consultation, diagnostics, prescriptions	Product privacy controls	Conformance evidence outside reviewed scope	Service catalogues	Commercial services [9, 10]
Arogga	E-pharmacy and fulfilment	Transactional account controls	Conformance evidence outside reviewed scope	Large commercial catalogue	Commercial service [11]
Medora	Patient-held workflow plus research test bed	Purpose, provider, scope, validity, revocation, audit	Registry-bounded actions; source-linked drafts; local-first paths	7,389 drugs; 67,001 brands	Live demo; MIT; archived v1.0.2 [12]

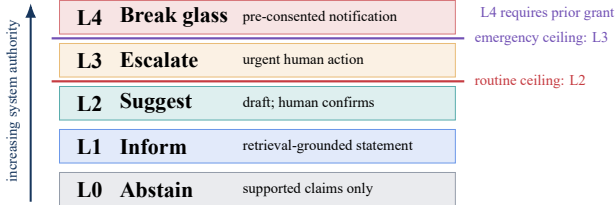


Status is written inside every node; border styles reinforce, but do not replace, the label.

Figure 2. Medora composes five artifacts around a deny-by-default consent engine. Deployed means running in the platform; measured means archived results exist; specified means design complete and scheduled for implementation; planned means a registered protocol scheduled for execution. Each node prints its status explicitly.

Table 2. Arohon authority tiers.

Tier / mode	Human gate	Constraint
L0 Abstain	none	supported claims only
L1 Inform	reads	source-linked
L2 Suggest	confirms	before effect
L3 Escalate	acts	recall-first
L4 Break glass	notified	prior revocable grant

**Figure 3.** Authority rises from abstention to a pre-consented break-glass notification within risk-specific ceilings. Risk policy determines the tier independently of model confidence.

tion rather than a clinical guideline.

Two traces make the specification testable. “Chest pain and difficulty breathing” maps to the cardiac risk class, pre-empts the model, and proposes a bilingual L3 takeover with a one-tap emergency action; dismissal becomes a labelled false-positive event. “I do not want to live” maps to the separate self-harm class, remains at L3, and surfaces human support while preserving a manual choice. Both traces are finalist implementation targets; the demo uses a mock number and a simulated emergency action.

3.2 Architecture and infrastructure

Every AI request passes through one control point. Medora comprises a Next.js PWA, a FastAPI core, a separate FastAPI OCR service, and PostgreSQL. Its core exposes 21 assistive-AI endpoints in nine groups and authenticates the actor and checks role, ownership or care relationship, appointment state, and consent. External text, live audio, and cloud OCR are different recipients with different grants. Local Whisper, YOLO, and PaddleOCR remain within the deployment, so local mode stays local.

Learned components perceive and draft; deterministic code decides authority. Speech transcription, prescription-region detection, OCR, and hosted text generation use learned models. Consent resolution, redaction, pseudonymi-

sation, intent normalisation, role-scoped route lookup, specialty and medicine matching, emergency rules, dosage grammar, and structured parsing remain inspectable code paths. The orchestrator sanitises every level of its input and replaces stable subject identifiers with a random per-request correlation ID. It then validates the output against a schema and records only provider and latency metadata; raw prompts and findings stay out of the default logs. Chorui’s immutable registry contains 24 entries over 23 intents: 11 patient, 11 clinician, and two shared; administrative routes remain outside its scope. Table 3 makes training, inference, evaluation, and status explicit.

Human authority is structural. Summaries are read-only and receive source references from the backend after validation, constraining source references to supplied identifiers. Specialty search returns candidates beside manual browsing. Consultation outputs pre-fill forms a doctor edits and signs. OCR rows carry `authoritative_writeback=false` and require row-level confirmation. Medora’s role is navigation, reviewed drafting, and non-authoritative extraction.

Administrative authority is scoped, not global. The deployed platform authorises administration through a single role; the specified stewardship layer binds each administrator to an affiliation, so authority becomes a scope multiplied by a permission set. A central administrator keeps platform-wide control and is the only role that can create other administrators. An organisation administrator sees only the doctors, appointments, and records of its own hospital or clinic, and a facility administrator is limited to one branch. A separate data-protection role can read consent and audit records but holds no operational control. Because the ordinance makes a hospital the data controller for its patients, the organisation scope makes that boundary observable and testable. Every privileged action joins an administrative audit beside the existing patient-access log, destructive actions require a second administrator, and emergency elevation follows the L4 break-glass rule and is logged. The same audit already lets a patient inspect who accessed a record and extends toward the ordinance’s export and erasure rights. Scoped roles, the affiliation model, and time-boxed delegated grants are specified over the deployed flat-

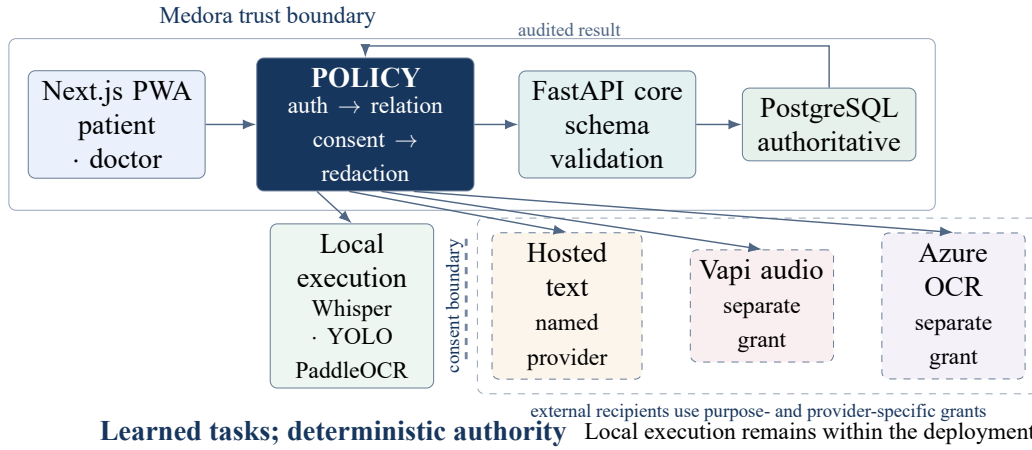


Figure 4. Control flow and trust boundaries. Every inference path passes through the policy chokepoint; PostgreSQL remains authoritative after reconnect and external recipients are separately consented.

Table 3. AI and algorithm inventory. “Deployed option” means the path exists in the live stack and may require configuration or a separate grant; planned rows identify registered evaluation work.

Task	Model / algorithm	Data source	Train / infer	Status	Evaluation	Safety boundary
Dictation	faster-whisper, Whisper small	consented live clinician audio	pretrained; local inference	Deployed option	workflow and route tests	local processing; editable draft
Live voice	Vapi tool bridge	separately consented audio	provider inference	Deployed option	route and guard tests	separate recipient grant; same registry
Region detection	project-trained YOLO ONNX	private student-project prescription images	local training; local inference	Deployed prototype	workflow evaluation; clinical study scheduled	regions remain non-authoritative
OCR	PaddleOCR / Azure Document Intelligence	prescription or report image	pretrained; local or cloud inference	Deployed options	review-gated prototype; clinical study scheduled	row confirmation; cloud grant separate
Text drafting	Groq GPT-OSS-120B default; Gemini, Cerebras, mock	redacted task payload and authorised record subset	provider inference	Deployed options	12 source-accounting fixtures; schema tests	policy precedes model; backend attaches sources
Drug matching	trigram, edit distance, dosage grammar	Akkhor corpus	rules; local inference	Deployed	reproducible corpus counts	candidates only; clinician confirms OCR rows
Navigation	registry, synonyms, fuzzy/similarity matching	curated intents, specialties, red-flag rules	rules; local inference	Measured	30 reviewed fixtures; 7/7 emergency labels retained; 5 conservative escalations	red flags pre-empt generation; manual browse remains
PHI redaction	bilingual patterns and pseudonymisation	134 synthetic production-path cases	rules; local inference	Measured	P 94.7%, R 75.5%, over-redaction 3.2%	measured baseline informs recall upgrade
PHI span recognition	MuRIL / XLM-R; BanglaBERT research comparator [13, 14, 15]	planned 8k–15k synthetic Bengali, English, romanised cases; existing 134 held out	token-classifier fine-tuning planned	Planned	rules vs model vs union; P/R/F1, over-redaction, CPU latency, 3 seeds	recall-first threshold; ONNX feature flag; licence gate
Maya gate	TigerLLM-1B candidate with LoRA [16]	redistributable, de-identified Bangla dialogue; corpus selection scheduled	fine-tuning planned	Planned	escalation sensitivity on reviewed red flags plus benign controls and CIs	admission gate precedes deployment; tier remains policy-set

admin controls.

Transactional and database controls cover non-model execution paths. Appointment creation locks the idempotency key and slot, then commits

the appointment, replay result, audit entry, and outbox event together; a unique active-slot constraint is the final guard. Across 30 fresh-slot repetitions at each of 2, 10, and 50 concurrent attempts, all 90 trials preserved exactly one commit. Trans-

action p95 was 38.8/580.8/1,571.5 ms and outbox-processing p95 was 67.7/696.0/1,520.9 ms. These fixture results establish contention behavior under the reported test envelope. Events publish after commit and reconnecting clients re-query PostgreSQL. A release audit also found anonymous database grants with row-level security disabled. The release revoked the grants, enabled deny-by-default RLS, restricted direct subscriptions to a trigger-maintained feed, pinned trigger search paths, and added negative access tests. The archived v1.0.2 receipt records passing backend, AI-service, integration, security, container, and frontend gates; all 12 English/Bengali browser specifications executed their authenticated journeys [12].

4. Evaluation and research plan

4.1 Shimana: planned consent–utility frontier

Shimana asks a measurable question: how does task utility change as disclosure is broadened? A fixed bilingual task set will run under five configurations: local only (L), local plus Akkhor (L+K), a redacted record subset (L+K+R), a hosted model under a matching grant (L+K+R+H), and an offline, de-identified unrestricted research control (U). Utility comprises specialty-navigation agreement, summary source-accounting, and medicine-match precision. Privacy exposure is undetected PHI spans per 1,000 requests. The reporter will emit paired results, bootstrap confidence intervals, the non-dominated set, and a knee estimate.

4.2 Lokkhon: measured safety baselines

Lokkhon publishes sample size beside every metric. Its status is measured for axes A–C, pilot (n=4) for D, and planned for E. Axis A tests red-flag escalation; B tests indirect prompt injection; C tests identifier leakage and over-redaction; D tests romanised code-mixing; and E will measure calibrated abstention with a risk-coverage curve. The current harness archives machine-readable cases, exclusion and outcome accounting, mock/live separation, and bootstrap reporters.

The current results establish containment baselines. The privacy suite contains 134 synthetic production-path cases and yields 94.7% span precision, 75.5% recall, 3.2% over-redaction, and 43

documented boundary notes. Extended rules for the leaking classes (unlabelled and cued names, obfuscated emails, textual dates, bare IDs, addresses) now redact every annotated span on this development set at 96.9% precision and 2.4% over-redaction. Since those rules were tuned on that set, a held-out probe of novel identifiers is the honest test: structured and cued classes generalise; the residual narrows to unlabelled, unseen names. The reviewed navigation set contains 30 fixtures: all seven emergency labels were recorded on both scored paths, with complete emergency-label retention and five conservative escalations. A licensed reviewer changed two authored Bengali labels to emergency before the rules were frozen. Twelve summary fixtures account for every backend-supplied source under the deterministic mock. These constructed sets establish a reproducible baseline for broader population evaluation.

Prescription OCR remains a review-gated prototype. The handwriting pipeline supports workflow integration while clinical-accuracy evaluation awaits an independently adjudicated corpus. The student-project image set stays private, since its permission covered project use rather than public redistribution. Every extracted row is non-authoritative and needs clinician confirmation, so we report workflow status and the next validation step, not an accuracy percentage.

4.3 Planned learned PHI recognition

The first post-submission build advances the measured redaction baseline. The residual after the extended rules is unlabelled, previously unseen names in both scripts, which pattern rules cannot generalise to; recognising unseen names is exactly what a learned model does. A reproducible generator will create 8,000–15,000 Bengali, English, and romanised sentences, including hard negatives. Each sentence is BIO-tagged for names, clinicians, phones, NIDs, addresses, dates, ages at least 90, hospitals, email, and medical-record numbers. MuRIL and XLM-R are deployment candidates; BanglaBERT is retained only as a research comparator because its published licence is non-commercial. The existing 134 cases remain untouched as a held-out test. Across three seeds, rules, each model, and a recall-first union will report span precision, recall, F1, false-redaction rate, bootstrap intervals, and CPU

Consent configurations

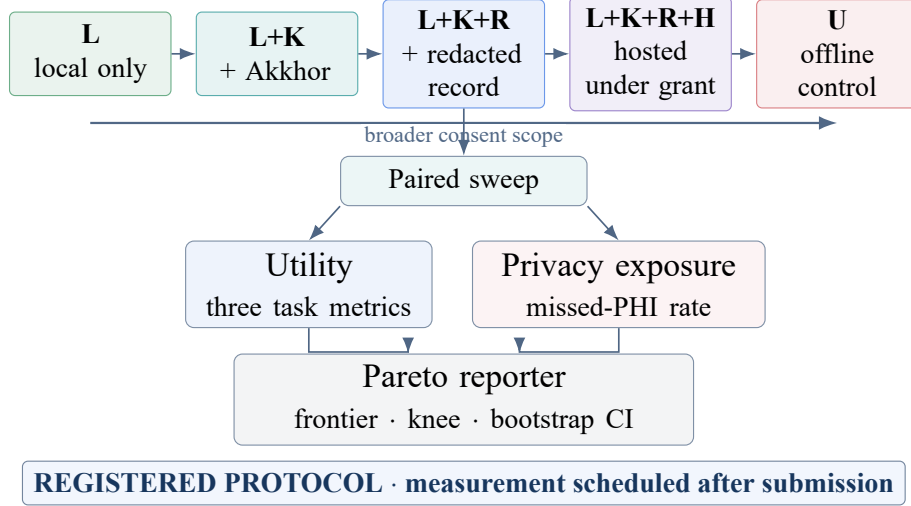


Figure 5. Registered Shimana experiment. The diagram fixes interventions, paired metrics, and reporting outputs; execution is scheduled after submission.

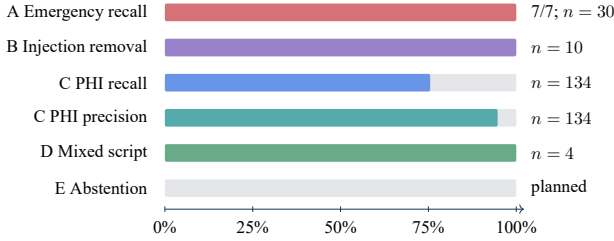


Figure 6. Current Lokkhon baselines. Axis D is an explicitly labelled four-case pilot; axis E is scheduled for the next benchmark release.

latency. Only a licence-cleared ONNX model that improves recall within the predefined over-redaction limit can enter the platform behind a feature flag. This registered build and evaluation plan defines the release gate.

4.4 Maya: planned model-admission gate

Maya turns escalation sensitivity into a deployment admission test. The hypothesis is that non-emergency dialogue can teach a small model to reassure when escalation is required. TigerLLM-1B is the current candidate; final selection follows the licence and corpus review. LoRA tuning would use a redistributable, de-identified Bangla dialogue corpus after translation, licence, and ethics review. The frozen comparison will score the base and tuned model on the clinician-reviewed red-flag set plus 25–30 benign controls. Primary outcome is first-sentence escalation sensitivity with a confidence interval; benign false-alarm rate detects trivial always-alert behaviour.

Self-harm cases use a separate agency-preserving rubric. Deployment admission requires passing the gate, while Arohon continues to select the tier. Corpus selection, adapter training, and archival execution are scheduled, so the paper reports the protocol.

5. Risk management and validation scope

The team tracks risk as an engineering artifact. Table 4 distinguishes deployed controls from research commitments and pairs each initial risk with a residual band and next validation step. Consent revocation governs future disclosure; 75.5% recall is the baseline for the planned union recogniser; and seven reviewed emergencies anchor expansion of clinical coverage.

6. Akkhor, feasibility, and distribution

Akkhor is reusable Bangladesh drug infrastructure rather than an application feature. A reproducible script reconciles five published datasets into immutable canonical drug identities keyed by generic, strength, and form; commercial brands map many-to-one; and a trigram index supports typo-tolerant lookup. The loaded release contains 71,795 source rows, 7,389 canonical drugs, 67,001 brand entries, 74,390 search terms, 5,242 distinct generics, and 52,117 distinct brands. Provenance is preserved per source. Current validation covers reproducible identity construction; regulator review and indication validation are future work.

Table 4. Risk register. Evidence refers to the current release; P denotes a planned 2.0 control. H/M/L are qualitative impact or residual-risk bands, not probabilities.

Risk (initial → residual)	Control	Current evidence / next step
Clinical-advice accuracy (H→M)	deterministic red flags; drafts only; P Arohon	30 reviewed fixtures; P tier logs
PHI exposure (H→M)	deny-by-default grant; rule redaction; P union span recogniser	134 cases; 75.5% recall baseline
Consent alignment (H→L-M)	purpose, provider, scope, version, validity, revocation	unit and integration tests
Administrative privilege/scope (M→L-M)	P scoped roles; two-person destructive actions; admin audit; logged L4 break-glass	deployed flat admin; affiliation scoping specified
Indirect prompt injection (H→M)	typed schema; out-of-band registry and policy	10/10 cases removed
Source grounding (H→M)	source attachment; Akkhor grounding; L0 abstention	12 summary fixtures; axis E scheduled
Escalation sensitivity (H→M)	model-independent emergency tier; P Maya admission gate	registered protocol
OCR extraction quality (H→M)	non-authoritative output; mandatory confirmation	clinical evaluation scheduled
Provider continuity/retention (M→M)	provider abstraction; local paths; retention manifest	manifest present
Emergency-service continuity (H→M)	time-aware registry; manual alternatives	P health checks
Cost scaling (M→L-M)	deterministic routing; caching; batch; spend caps	planning envelope
Clinician adoption (H→H)	editable drafts; patients free; physician-workflow pilot	pilot scheduled
Banglish/code-mix robustness (M→M)	transliteration-aware cases and encoders	four-case pilot; expansion scheduled
Dataset licence compatibility (H→M)	per-source provenance; research/deployment licence gate	corpus selection scheduled
Clinician review capacity (M→M)	publish adjudicated samples; version benchmark	reviewer capacity determines scale

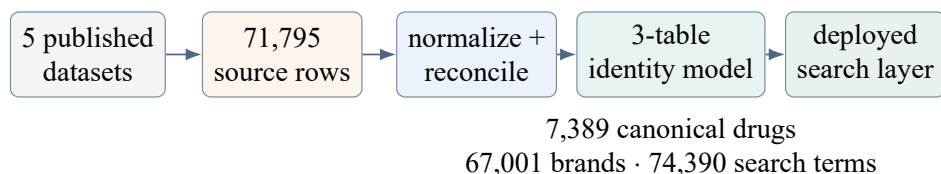


Figure 7. Akkhor turns five source datasets into canonical identity and a deployed platform search layer. The counts are generated from the current loaded corpus.

Akkhor stays free; revenue comes from workflow, hosting, and conformance, never from the open assets. Application code is MIT-licensed and the compiled medicine corpus is CC

BY 4.0, so neither Akkhor nor patient access is ever a paid line. The open core instead supports four service streams. A chamber and clinic subscription sells the clinician workflow; the value

Table 5. Monthly infrastructure scenarios in US dollars. The 1,000-MAU column starts from the repository deployment estimate; higher Azure columns use linear scaling as a conservative stress assumption. Text assumes three calls/MAU, 1,000 input and 250 output tokens/call at current Groq GPT-OSS-120B rates [17].

Line item	1k MAU	10k MAU	100k MAU	Basis / unresolved variable
Backend containers and registry	20–35	200–350	2,000–3,500	current FastAPI container plus Azure registry estimate
OCR: local container + cloud	30–100	300–1,000	3,000–10,000	local PaddleOCR/YOLO plus usage-dependent Document Intelligence
PostgreSQL and auth	25	25–75	25–125	Supabase Pro starts at US\$25; compute upgrades are usage-dependent [18]
Groq text inference	0.90	9.00	90.00	uncached input US\$0.15/M; output US\$0.60/M
Cached or batch text case	0.45–0.68	4.50–6.75	45.00–67.50	eligible-text alternative shown separately from totals
Monitoring	0–10	10–50	50–200	telemetry and retention determine the tier
Staffed support	Pilot	Pilot	Pilot	tickets and clinician onboarding measured in pilot
Subtotal before support	75.90–170.90	544–1,484	5,165–13,915	tax, FX, optional Vapi audio, and support listed separately
With 20% contingency	91.08–205.08	652.80–1,780.80	6,198–16,698	contingency applies to priced rows only
Infrastructure / paying doctor	1.82–4.10	1.31–3.56	1.24–3.34	assumes 20 active patients per paying doctor; excludes support

proposition is less clerical time, with a design target of two minutes saved across 30 daily consultations. An institution stream sells on-premise deployment on a privacy premium with support. A conformance stream offers Lokkhon-based safety testing and consent-conformance review to other applications that must demonstrate PDPO handling. This reuses the benchmark and never touches the free dataset. The same in-boundary stack also reaches adjacent public-sector and financial deployments, the market opened by the 2026 government cautions noted in Section 1. Table 6 lists the streams and their pilot pricing hypotheses; willingness to pay is not yet measured.

The market is sized conservatively and bottom-up. Statista projects the Bangladesh digital-health market at US\$849.3m by 2029 [22]; Medora addresses the workflow and compliance slice rather than the consultation market itself. The serviceable base is private practice: the BMDC listed 134,568 registered physicians in 2024, of whom roughly 98,000 hold current renewal [23]. The obtainable pilot share is deliberately narrow, the KUET-network chambers in Khulna on the order of a few hundred physicians at the hypothesised BDT 1,200 per month, and any scale claim waits on measured retention and conversion from that pilot.

Distribution advances by evidence gate, not by calendar. Phase 0 runs the Khulna chamber pilot and measures time saved, onboarding cost, retention, and paid conversion. Phase 1 opens multi-seat clinics on validated pricing. Phase 2 adds institution on-premise deployments and the conformance service as the ordinance’s delayed enforce-

Table 6. Pilot subscription hypotheses; willingness-to-pay measurement is scheduled.

Tier	Price	Included scope
Patient	Free	records, consent, booking, navigation
Akkhor	Free	open dataset (CC BY 4.0); never a paid line
Chamber	BDT 1,200 per month	one physician workflow
Clinic	BDT 900 per seat per month	shared clinic administration
Institution	annual quote	on-premise deployment and support
Conformance	engagement quote	Lokkhon testing and consent-conformance review

ment provisions commence. Phase 3 extends the in-boundary stack to adjacent regulated sectors.

The cost model identifies its next measurement variables. Azure Container Apps can scale to zero and includes a monthly consumption allowance [19]; Supabase Pro includes up to 100,000 monthly active users but has separate database, storage, and egress limits. Groq prompt caching can halve cached-input cost and asynchronous batches can reduce eligible non-interactive work by 50%, but neither removes database, OCR, or support cost [17]. The next budgeting evidence is therefore telemetry plus staffed support time. The PDPO may increase demand for demonstrable safeguards; any certification claim would follow an accredited scheme.

7. Roadmap, team, and conclusion

The finalist plan starts with the measured redaction baseline. First, build and evaluate the planned PHI span recogniser against the untouched 134-case holdout, publishing the generator, model cards, seed-level results, and rule/model/union comparison. Second, add Arohon enums, ceiling configuration, tier logs, and fixtures; publish Lokkhon v0.1 at the current adjudicated sample size; and run an initial Shimana sweep. Third, run Maya after corpus and model licensing review, freeze machine-readable results, and add the L3 emergency interface. Each stage advances after its release gate is satisfied.

At submission the five artifacts sit at different evidence levels: Arohon specified, Lokkhon measured (axes A–C; D a four-case pilot; E planned), Shimana and Maya planned, and Akkhor deployed. The learned PHI upgrade is scheduled for training, evaluation, and a release gate.

Roles match the evidence path. Sarwad Hasan Siddiqui leads architecture, backend, AI/ML, deployment, benchmarking, and paper integration. Adiba Tahsin leads the PWA, bilingual consent and emergency UX, figures, and demo validation. Kazi Saeed Alam supervises method, evaluation, and research communication. The underlying v1.0.2 software and evaluation harness are archived on Zenodo [12] and described in a manuscript submitted to SoftwareX; this competition paper reframes and extends that artifact rather than duplicating its claims.

Medora’s contribution is simple to state: control who and what an AI may touch, and keep a human in charge of the result. The evidence so far, 134 privacy cases, 30 reviewed navigation fixtures, 12 grounded summaries, and a 7,389-drug identity layer, shows the controls run end to end and gives a baseline to grow from. Shimana and Maya, wider clinician review, and higher de-identification recall are the next evidence, and the AI stays bounded as that evidence grows.

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